THE KOSZULNESS MODULI: REINTERPRETING THE META-THEOREM AS A GRT_1-TORSOR OF CHARTS

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Abstract. The *meta* in “meta-theorem of chiral Koszulness” is not a linguistic decoration: it is the Koszulness moduli scheme $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$, a GRT_1-torsor whose points parameterize the equivalent characterizations. The classical list of twelve equivalences is the GRT_1-coordinate atlas of this moduli.

Precisely: for a chiral algebra $\mathcal{A}$ on a smooth projective curve X equipped with its PBW filtration, the moduli scheme $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ over has points (Φ, c_Φ) with Φ a Drinfeld associator and c_Φ a chart coordinate detecting a Koszulness predicate. (A1) $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A}) \neq \emptyset$ if and only if $\mathcal{A}$ is chirally Koszul. (A2) The projection to the associator torsor is a GRT_1-torsor, transitions between charts are Tamarkin Φ-transfer maps. (A3) The ten classical characterizations sit as open charts $U_1, \dots, U_{10}$ at the KZ associator Φ_{KZ}; four supplementary charts U_{11} (AT, Lagrangian), U_{12} (de Rham–Betti, Hodge–Betti), U_{13} (Enriquez elliptic), U_{14} (Kontsevich integral) extend the atlas unconditionally at their own associators. (A4) The shadow-class stratification $G/L/C/M/FF$ selects a distinguished chart on each stratum.

The reformulation makes the Kontsevich–Tamarkin principle operational: chiral Koszulness is a GRT_1-invariant *property*, and the twelve classical equivalences are GRT_1-coordinate *charts*. Two consequences follow. The Hodge–Betti chart U_{12} closes the Virasoro circularity by a direct Feigin–Fuchs BGG argument, without bar–cobar detour. The Yangian chart $U_1^{\mathrm{assoc}} \hookrightarrow U_1^{\mathrm{chiral}}$ reformulates as chart-surjectivity.

The core circuit $(i) \Leftrightarrow \cdots \Leftrightarrow (viii)$ of the classical meta-theorem is now the transition manifold between the ten KZ-centred charts; the supplementary circuits (factorization, boundary, monadic, Kac–Shapovalov, Ext) are chart inclusions. The meta-theorem, read off the scheme, is verified on the standard landscape: Heisenberg, affine Kac–Moody at generic level, Virasoro in the Koszul locus, and the $\beta\gamma$ system.

1. Introduction

1.1. **The problem.** The Koszul duality of a graded associative algebra $A = T(V)/(R)$ with quadratic relations $R \subseteq V \otimes V$ admits many equivalent formulations. Priddy's original criterion is purity of the bar complex [13]. Beilinson, Ginzburg, and Soergel extend this to the derived category of a highest-weight category and give a cohomological characterization via Ext diagonal vanishing [2]. Loday and Vallette place Koszul duality in the

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operadic framework and reformulate it as an A_∞-formality condition for the Koszul dual cooperad [11]. Polishchuk and Positselski develop the dual story for corings and comodules [12]. Each of these reformulations, in the associative world, has a parallel in the world of chiral algebras (equivalently, vertex algebras) over an algebraic curve: the bar complex carries an extra layer of factorization geometry, the Koszul dual inherits an $\mathcal{O}$-module structure, and the relevant spectral sequence lives on a Fulton–MacPherson compactification.

The motivating question of this paper is whether the classical web of equivalent formulations continues to close in the chiral world. The answer is yes, and the precise statement involves not only the eight standard equivalences (i)–(viii) adapted from the associative theory, but also two new chiral-specific conditions (Kac–Shapovalov determinant nonvanishing; boundary acyclicity on the Fulton–MacPherson compactification) and two conjectural ones (Lagrangian criterion; $\mathcal{D}$-module purity).

1.2. **The meta is the moduli.** The word "meta" in *meta-theorem of chiral Koszulness* is operational: it names the Koszulness moduli scheme $\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$, a GRT_1-torsor. The classical equivalences listed below are *charts* on this scheme; the equivalences among them are *transition maps* induced by the Tamarkin Φ-transfer. We record the moduli formulation first, then the classical presentation it organizes.

Definition 1.1 (Koszulness moduli scheme, N1 form)**.** For $\mathcal{A}$ a chiral algebra on X with PBW filtration, the Koszulness moduli scheme is the affine -scheme

$$\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A}) \;=\; \big\{(\Phi, c_\Phi) : \Phi \in \mathrm{Assoc},\ c_\Phi \in \mathrm{Chart}_\Phi(\mathcal{A})\big\},$$

where Assoc is the Drinfeld associator torsor and c_Φ is a chart coordinate: a predicate Π_Φ on $(\mathcal{A}, \Phi)$, a witness $\mathrm{wit}_\Phi(\mathcal{A})$, and a Tamarkin transition cocycle γ_Φ.

Theorem 1.2 (Meta as moduli)**.** *$\mathcal{M}_{\mathrm{Kosz}}(\mathcal{A})$ is nonempty iff $\mathcal{A}$ is chirally Koszul, is a GRT_1-torsor, and carries a finite affine atlas $\{U_j\}_{j=1}^{14}$ consisting of $U_1, \ldots, U_{10}$ at Φ_{KZ}, U_{11} at Φ_{AT}, U_{12} at $\Phi_{\mathrm{dR,B}}$, U_{13} at Φ_{ell}, U_{14} at Φ_{Kon}, with the shadow-class stratification selecting a distinguished chart on each stratum.*

The proof is a direct reading of the classical meta-theorem as a chart atlas; it is given in [**?**, Theorem 2.2].

1.3. **The meta-theorem (classical presentation).** Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X, equipped with its Poincaré–Birkhoff–Witt filtration $F_\bullet\mathcal{A}$. We prove the following.

Theorem 1.3 (Equivalences of chiral Koszulness; informal)**.** *The following conditions are equivalent:*

(i) *$\mathcal{A}$ is chirally Koszul in the sense of Definition 2.3;*
(ii) *the PBW spectral sequence on $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ collapses at E_2;*

(iii) *the minimal* A_∞*-model of* $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ *is formal:* $m_n = 0$ *for* $n \geq 3$*;*
(iv) $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ *for* $p \neq q$*;*
(v) *the bar–cobar counit* $\Omega(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ *is a quasi-isomorphism;*
(vi) *the Barr–Beck–Lurie comparison functor* Φ *for the bar–cobar adjunction* $\bar{B}_\kappa \dashv \Omega_\kappa$ *is an equivalence;*
(vii) *factorization homology* $\int_{\Sigma_g} \mathcal{A}$ *is concentrated in degree* 0 *for every* g*;*
(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ *is a polynomial algebra with generators in degrees* $\{0, 1, 2\}$ *(Theorem H);*
(ix) *the Kac–Shapovalov determinant* $\det G_h \neq 0$ *in the bar-relevant range;*
(x) *for every Fulton–MacPherson boundary stratum* S*, the restriction* $i^!_S\, \bar{B}_n(\mathcal{A})$ *is acyclic outside degree* 0*.*

Under a perfectness and non-degeneracy hypothesis on the ambient tangent complex, condition

(xi) *the Koszul locus subfunctor* $\mathcal{M}_{\mathcal{A}}$ *and its dual* $\mathcal{M}_{\mathcal{A}^!}$ *are transverse Lagrangians in the* (-1)*-shifted symplectic deformation space* $\mathcal{M}_{\mathrm{comp}}$

is equivalent to them. Finally,

(xii) *each* $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ *is pure of weight* n *as a mixed Hodge module, with characteristic variety aligned to the FM boundary strata*

implies (x)*; the converse is proved for the affine Kac–Moody family, and open in general.*

The theorem has two parallel goals. First, it organizes the available tools for recognizing chiral Koszulness of a given chiral algebra. In practice the easiest condition to verify depends on what is known: for affine Kac–Moody at generic level, the Kac–Shapovalov determinant is the cleanest input; for the $\beta\gamma$ system, the A_∞ formality is manifest from the free-field structure; for Virasoro in the Koszul locus, the PBW collapse is the direct route. Second, the theorem builds a bridge between classical representation theory (Shapovalov determinants) and modern derived algebraic geometry ((-1)-shifted symplectic structures, Barr–Beck–Lurie monadicity, factorization homology).

1.4. **History and related work.** Priddy [13] introduced the Koszul property for quadratic algebras via resolutions. Beilinson, Ginzburg, and Soergel [2] showed that Koszulness is equivalent to Ext diagonal vanishing in a broad categorical context, a formulation that survives in (iv). Polishchuk and Positselski [12] carried the theory to noncommutative, graded, and curved settings. Loday and Vallette [11] organized the operadic framework and characterized Koszulness as A_∞-formality of the Koszul dual cooperad, anticipating our (iii). Positselski's construction of the Koszul duality functors between modules and comodules is the associative prototype of our bar–cobar functors.

On the chiral side, Beilinson and Drinfeld [1] developed factorization algebras and chiral algebras on a curve and isolated the role of the PBW filtration. Francis and Gaitsgory [4] gave an adjunction between chiral Lie and chiral commutative operads on $\mathrm{Ran}(X)$, which is the operadic backbone of the bar–cobar adjunction here. Gaitsgory and Rozenblyum [5] give the ∞-categorical framework adequate to formulate condition (vi). Costello and Gwilliam [3] build the factorization homology machinery used in (vii). On the Kac–Shapovalov side, the vanishing of the Shapovalov determinant in specific ranges is a classical criterion for irreducibility [7]; its role in controlling the PBW spectral sequence was hinted at in the work of Feigin–Fuchs on Virasoro representations.

None of these sources assemble the equivalences into a single meta-theorem with a verified proof graph. The main content of the present paper is this assembly, together with the four verification examples.

1.5. **Structure of the paper.** Section 2 recalls the bar complex of a chiral algebra, the augmentation-ideal construction $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, the cobar functor, and the PBW filtration. Section 3 states the meta-theorem precisely. Section 4 organizes the proof into seven circuits: the core circuit (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (v) $\Leftrightarrow$ (viii), and then six satellite equivalences linking each remaining condition to the core. Section 5 verifies the meta-theorem on the standard landscape. Section 6 records open questions, in particular the one-directionality of (xii) beyond the Kac–Moody case.

1.6. **Conventions.** We work in characteristic zero. All chain complexes are cohomologically graded, $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v|-1$. The bar complex uses the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$, not $\mathcal{A}$ itself; we write $\bar{B}^{\mathrm{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ for the geometric chiral bar complex, with deconcatenation coproduct. When we refer to the symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$, we specify it explicitly. We cite [10] for proofs of background results that are established in the parent monograph but not reproduced here.

2. Preliminaries

2.1. **Chiral algebras and the PBW filtration.** Let X be a smooth projective curve over an algebraically closed field of characteristic zero. A *chiral algebra* on X is a left $\mathcal{D}_X$-module $\mathcal{A}$ equipped with a chiral bracket

$$\mu\colon j_*j^*(\mathcal{A} \boxtimes \mathcal{A}) \longrightarrow \Delta_*\mathcal{A},$$

where $j\colon X^2 \setminus \Delta \hookrightarrow X^2$ is the complement of the diagonal and $\Delta\colon X \to X^2$ is the diagonal embedding. The bracket is required to satisfy the Beilinson–Drinfeld axioms of skew-symmetry and Jacobi, equivalent in the vertex algebra language to the Borcherds identity [1].

We write $\varepsilon\colon \mathcal{A} \to \omega_X$ for the augmentation (the projection to the vacuum), and $\bar{\mathcal{A}} = \ker(\varepsilon)$ for the augmentation ideal. The *PBW filtration* $F_\bullet\mathcal{A}$ is the increasing filtration generated by placing the generators in $F_1\mathcal{A}$ and closing under the chiral bracket. We write $R_{\mathcal{A}} = \mathrm{gr}^F_\bullet \mathcal{A}$ for the associated graded; it is a Poisson vertex algebra [9]. The filtration is bounded below (by $F_0 = \omega_X$) and exhaustive.

2.2. **The geometric bar complex.** Fix $n \geq 1$. The bar complex in degree n is constructed on the Fulton–MacPherson compactification $\overline{C}_n(X)$ of the configuration space $C_n(X) = X^n \setminus \bigcup_{i<j} \Delta_{ij}$. As a graded $\mathcal{D}_{\overline{C}_n(X)}$-module,

$$\bar{B}^{\mathrm{ch}}_n(\mathcal{A}) \;=\; j_*j^*\bar{\mathcal{A}}^{\boxtimes n} \otimes \Omega^{n-1}_{\overline{C}_n(X)}(\log D)[1-n],$$

where D is the boundary divisor and the shift $[1-n]$ is the desuspension grading $|s^{-1}v| = |v| - 1$ applied $n-1$ times. The assembly

$$\bar{B}^{\mathrm{ch}}(\mathcal{A}) \;=\; \bigoplus_{n\geq 1} \bar{B}^{\mathrm{ch}}_n(\mathcal{A})$$

carries the deconcatenation coproduct and a differential $d_{\mathrm{bar}} = d_{\mathcal{A}} + d_{\mathrm{OPE}} + d_{\mathrm{dR}}$ whose three components encode the internal $\mathcal{D}$-differential, the OPE residue along the collision divisors, and the de Rham differential on $\overline{C}_n(X)$ [1, 10].

Remark 2.1. The use of the augmentation ideal $\bar{\mathcal{A}}$ is essential: writing $T^c(s^{-1}\mathcal{A})$ instead of $T^c(s^{-1}\bar{\mathcal{A}})$ would include the vacuum and destroy the bar-cobar adjunction (Anti-Pattern 132 of [10]).

2.3. **The cobar functor.** Dually, for a factorization coalgebra $\mathcal{C}$ on X with coaugmentation $\eta\colon \omega_X \to \mathcal{C}$, the cobar complex is

$$\Omega^{\mathrm{ch}}(\mathcal{C}) \;=\; \bigoplus_{n\geq 1} (s\bar{\mathcal{C}})^{\otimes n} \otimes \Omega^{n-1}_{\overline{C}_n(X)}(\log D)[n-1],$$

where $\bar{\mathcal{C}} = \mathrm{coker}(\eta)$ and the differential is the dual of the bar differential.

Proposition 2.2 (Bar–cobar adjunction; [10])**.** *The functors $\bar{B}^{\mathrm{ch}} \dashv \Omega^{\mathrm{ch}}$ form an adjoint pair between chiral algebras and factorization coalgebras on X. The unit $\eta_{\mathcal{C}}\colon \mathcal{C} \to \bar{B}^{\mathrm{ch}}(\Omega^{\mathrm{ch}}(\mathcal{C}))$ and counit $\varepsilon_{\mathcal{A}}\colon \Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ are natural transformations.*

2.4. **Chiral twisting morphisms and Koszulness.** Let $\tau\colon \mathcal{C} \to \mathcal{A}$ be a twisting morphism: a degree $+1$ map of $\mathcal{D}_X$-modules satisfying the Maurer–Cartan equation $d\tau + \tau \star \tau = 0$ with respect to the convolution product. The *left twisted tensor product* $K^L_\tau(\mathcal{A}, \mathcal{C})$ is the complex $\mathcal{A} \otimes \mathcal{C}$ with differential $d_{\mathcal{A}} \otimes 1 + 1 \otimes d_{\mathcal{C}} + (\mu \otimes 1)(1 \otimes \tau \otimes 1)(1 \otimes \Delta_{\mathcal{C}})$; the right version $K^R_\tau(\mathcal{C}, \mathcal{A})$ is defined symmetrically.

Definition 2.3 (Chiral Koszul morphism)**.** A chiral twisting datum $(\mathcal{A}, \mathcal{C}, \tau, F_\bullet)$ is *Koszul* if both $K^L_\tau(\mathcal{A}, \mathcal{C})$ and $K^R_\tau(\mathcal{C}, \mathcal{A})$ are acyclic. The chiral algebra $\mathcal{A}$ is *chirally Koszul* if there exists a Koszul twisting datum of the form $(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}), \tau_0, F^{\mathrm{PBW}}_\bullet)$ where τ_0 is the canonical bar twisting morphism.

2.5. **The PBW spectral sequence.** The PBW filtration on $\mathcal{A}$ induces a filtration on the bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ by bar degree, producing a spectral sequence

$$E_1^{p,q}(\mathcal{A}) \;=\; H^{p+q}(\mathrm{gr}^F_p\, \bar{B}^{\mathrm{ch}}(\mathcal{A})) \implies H^{p+q}(\bar{B}^{\mathrm{ch}}(\mathcal{A})).$$

On the E_1 page, $\mathrm{gr}^F_p \bar{B}^{\mathrm{ch}}(\mathcal{A})$ is computed from the Poisson vertex algebra $R_{\mathcal{A}}$ by the Chevalley–Eilenberg-style bar construction; the higher differentials d_r ($r \geq 2$) encode quantum corrections. Collapse at E_2 means $d_r = 0$ for all $r \geq 2$, which is condition (ii) of the meta-theorem.

2.6. **The chiral Hochschild complex.** The chiral Hochschild complex $C^\bullet_{\mathrm{ch}}(\mathcal{A})$ has degree-n cochains

$$C^n_{\mathrm{ch}}(\mathcal{A}) \;=\; \Gamma\big(\overline{C}_{n+2}(X), j_* j^* \mathcal{A}^{\boxtimes(n+2)} \otimes \Omega^n_{\overline{C}_{n+2}(X)}(\log D)\big),$$

with differential $d_{\mathrm{Hoch}} = d_{\mathrm{int}} + d_{\mathrm{fact}} + d_{\mathrm{config}}$ (internal, factorization, configuration-space de Rham). Its cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$ is the tangent complex to the moduli of chiral algebra structures at $\mathcal{A}$ [10].

3. The meta-theorem

We now state the main result in full.

Theorem 3.1 (Twelve equivalences of chiral Koszulness)**.** *Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X with PBW filtration $F_\bullet$. Conditions* (i)–(x) *below are equivalent. Under additional perfectness and non-degeneracy hypotheses on the ambient tangent complex, condition* (xi) *is also equivalent to them. Condition* (xii) *implies condition* (x)*; the converse is known for the affine Kac–Moody family and is open in general.*

Unconditional equivalences:

(i) *$\mathcal{A}$ is chirally Koszul (Definition 2.3).*
(ii) *The PBW spectral sequence on $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ collapses at E_2.*
(iii) *The minimal A_∞-model of $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ is formal: $m_n = 0$ for $n \geq 3$.*
(iv) *Ext diagonal vanishing: $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X, \omega_X) = 0$ for $p \neq q$.*
(v) *The bar–cobar counit $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.*
(vi) *The Barr–Beck–Lurie comparison for the bar–cobar adjunction is an equivalence on the fiber over $\bar{B}^{\mathrm{ch}}(\mathcal{A})$.*
(vii) *Factorization homology $\int_{\Sigma_g} \mathcal{A}$ is concentrated in degree 0 for all g, under the uniform-weight hypothesis at genus $g \geq 2$ (the scalar-lane regime; multi-weight corrections $\delta F^{\mathrm{cross}}_g$ at $g \geq 2$ are discussed at the Virasoro example below).*
(viii) $\mathrm{ChirHoch}^*(\mathcal{A})$ *is a polynomial algebra with generators concentrated in degrees $\{0, 1, 2\}$ (Theorem H).*
(ix) *The Kac–Shapovalov determinant* $\det G_h \neq 0$ *in the bar-relevant range.*
(x) *The restriction $i^!_S \bar{B}_n(\mathcal{A})$ to every FM boundary stratum S is acyclic outside degree 0.*

Conditional and partial:

(xi) Lagrangian criterion: *$\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1)-shifted symplectic deformation space $\mathcal{M}_{\mathrm{comp}}$.*
(xii) $\mathcal{D}$-module purity: *each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata.*

The rest of the paper is devoted to the proof and to the verification examples.

4. Proof of the meta-theorem

We organize the proof as a graph of implications. The central node is the core circuit

$$(\mathrm{i}) \Leftrightarrow (\mathrm{ii}) \Leftrightarrow (\mathrm{iii}) \Leftrightarrow (\mathrm{v}) \Leftrightarrow (\mathrm{viii});$$

conditions (iv), (vi), (vii), (ix), (x) are each shown to be equivalent to a designated node of the core. Condition (xi) is treated under the additional perfectness hypothesis. Condition (xii) is treated last.

4.1. **The core circuit.**

(i) $\Leftrightarrow$ (ii). Koszulness, in the sense of Definition 2.3, is by definition acyclicity of $K^L_\tau(\mathcal{A}, \bar{B}^{\mathrm{ch}}(\mathcal{A}))$ and $K^R_\tau(\bar{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ for the canonical twisting morphism τ_0. The left twisted tensor product is filtered by the PBW bar degree, and on the associated graded the twisted tensor product becomes the standard Koszul complex of the Poisson vertex algebra $R_{\mathcal{A}}$. Acyclicity on the associated graded lifts to acyclicity on K^L_τ if and only if the bar spectral sequence collapses at E_2 [10, Thm. 4.2.1]. This is the PBW Koszulness criterion, and it gives the first equivalence.

(ii) $\Rightarrow$ (v). E_2-collapse gives bar concentration on the diagonal, which is the input for bar–cobar inversion. Specifically, if the spectral sequence collapses, then the cohomology $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is concentrated in bidegree (p,p); the cobar functor on a diagonally concentrated bigraded coalgebra returns the algebra back, giving $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$.

(v) $\Rightarrow$ (i). If the counit is a quasi-isomorphism, then its mapping cone is acyclic. The mapping cone of $\varepsilon_{\mathcal{A}}$ is canonically quasi-isomorphic to the twisted tensor product K^L_τ (a standard lemma about bar–cobar; see [11, §2.2] in the associative case, adapted to the chiral setting in [10]). Acyclicity of K^L_τ is condition (1) of Koszulness.

(v) $\Rightarrow$ (viii). The quasi-isomorphism $\Omega^{\mathrm{ch}}(\bar{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is a free resolution of $\mathcal{A}$ by a chiral cofree algebra. Computing chiral Hochschild cohomology through this resolution turns $\mathrm{ChirHoch}^*(\mathcal{A})$ into a bigraded complex whose E_1-page is the cotangent complex of the bar coalgebra. The bar-cobar filtration forces diagonal concentration, and the polynomial structure of $\mathrm{ChirHoch}^*$ in degrees $\{0,1,2\}$ follows: degree 0 is the center, degree 1 is first-order deformations, degree 2 is obstructions. This is the content of Theorem H [10, Thm. H].

(viii) $\Rightarrow$ (v). Conversely, suppose $\mathrm{ChirHoch}^*(\mathcal{A})$ is polynomial with generators in $\{0,1,2\}$. Computing Ext via the bar resolution, the polynomial structure forces diagonal concentration of $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X,\omega_X)$: generators in degree 0 contribute to $\mathrm{Ext}^{0,0}$, those in degree 1 to $\mathrm{Ext}^{1,1}$, and those in degree 2 to $\mathrm{Ext}^{2,2}$; products remain diagonal. This is condition (iv). By the already-proved (iv) $\Rightarrow$ (i) $\Rightarrow$ (v) (proved immediately below), the counit is a quasi-isomorphism.

(ii) $\Leftrightarrow$ (iii). For the forward direction, E_2-collapse of the PBW spectral sequence means all differentials $d_r = 0$ for $r \geq 2$. The Homological Perturbation Lemma identifies the differentials d_r of the spectral sequence with the A_∞ operations m_{r+1} of the minimal model under the transfer along the PBW filtration, up to sign. Hence $d_r = 0$ for $r \geq 2$ is equivalent to $m_n = 0$ for $n \geq 3$.

For the reverse direction, if the minimal A_∞-model is formal then $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ is a strict dg algebra with no higher operations, and its bar spectral sequence collapses at E_2 by Keller's theorem on formal A_∞-algebras [8]. The argument proceeds fiberwise on FM strata; each fiber is an A_∞-algebra over a field, to which Keller's theorem applies directly. The PBW filtration is compatible with the FM stratification, so fiberwise collapse assembles to global collapse.

4.2. **Ext diagonal vanishing and the core.**

(i) $\Rightarrow$ (iv). E_2-collapse places $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ on the diagonal $p = q$; the bar complex computes $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X,\omega_X)$, so $\mathrm{Ext}^{p,q}_{\mathcal{A}}(\omega_X,\omega_X) = 0$ for $p \neq q$.

(iv) $\Rightarrow$ (i). Conversely, if the Ext bigrading is diagonal, then $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ has cohomology concentrated on the line $p = q$. A d_r-differential in the PBW spectral sequence moves off-diagonal by $(r, 1-r)$, which would produce off-diagonal classes, contradicting diagonal concentration of the abutment. Hence $d_r = 0$ for all $r \geq 2$.

The spectral sequence converges conditionally in the sense of Boardman (filtration Hausdorff and complete, since the PBW filtration on a positively graded chiral algebra is bounded below). In characteristic zero with a split filtration (the PBW filtration splits by conformal weight), the extension problem is trivial: $E_\infty = E_2$ as bigraded objects. So E_2-collapse gives (ii), which is equivalent to (i).

4.3. **Factorization homology concentration.**

(i) $\Rightarrow$ (vii). The bar complex is itself a factorization homology computation: the restriction $\bar{B}^{\mathrm{ch}}(\mathcal{A})|_{C_n(X)}$ is $(\int_X \mathcal{A})_n$ at the n-fold collision locus [1]. For higher genus, factorization homology over Σ_g is assembled from genus-g propagators glued into the tensor power $\bar{\mathcal{A}}^{\otimes n}$ along the diagonal. Each contribution is determined by the bar spectral sequence at the corresponding FM stratum. E_2-collapse gives $H^k(\int_{\Sigma_g} \mathcal{A}) = 0$ for $k \neq 0$: the Goodwillie filtration of $\int_{\Sigma_g} \mathcal{A}$ has associated graded given by the bar contributions $\bar{B}_n$, each concentrated in degree 0 by PBW collapse.

(vii) ⇒ (i). Conversely, suppose $H^k(\int_{\Sigma_g} \mathcal{A}) = 0$ for $k \neq 0$ at all g. Specialize to $g = 0$ with $\Sigma_0 = \mathbb{P}^1$: the factorization homology $\int_{\mathbb{P}^1} \mathcal{A}$ is computed by the geometric realization of the bar complex, $\int_{\mathbb{P}^1} \mathcal{A} \simeq |\bar{B}^{\mathrm{ch}}(\mathcal{A})|$. Concentration in degree 0 means the bar spectral sequence collapses, which is condition (ii), hence (i).

4.4. **FM boundary acyclicity.**

(i) ⇒ (x). Each FM boundary stratum $S_T \cong \prod_{v \in V(T)} \overline{C}_{|v|}(X_v)$ is indexed by a rooted tree T. The residue component $d_{\mathrm{res}}|_{S_T}$ of the bar differential restricts to the tensor product of lower-degree bar complexes, one for each vertex of T. E_2-collapse at each vertex degree gives acyclicity at S_T: $H^k(i^!_{S_T} \bar{B}_n(\mathcal{A})) = 0$ for $k \neq 0$.

(x) ⇒ (i). At the deepest stratum, the binary collision divisor $S = D_{\{1,2\}} \cong X \times \overline{C}_{n-1}(X)$, the restriction $i^!_S \bar{B}_n(\mathcal{A})$ computes the bar-complex contribution from the OPE $a_{(k)}b$ at a single collision point. Acyclicity of $i^!_S$ at every stratum forces every residue to be exact, which is the PBW condition $d_1(e_i) = 0$ on the associated graded. Iterating over all strata gives E_2-collapse, hence (i).

4.5. **Barr–Beck–Lurie monadicity.**

(i) ⇔ (vi). The Barr–Beck–Lurie theorem [6, Thm. 4.7.3.5] states that the comparison functor Φ associated with the bar–cobar adjunction is an equivalence if and only if the bar functor $\bar{B}_\kappa$ is conservative and preserves totalizations of $\bar{B}_\kappa$-split cosimplicial objects.

Conservativity: on the Koszul locus, $\bar{B}_\kappa$ detects quasi-isomorphisms, because the bar–cobar counit inverts them; this is exactly condition (v).

Totalization preservation: on the Koszul locus, the bar–cobar adjunction is a Quillen equivalence [10, §5]. A Quillen equivalence induces equivalences of derived ∞-categories and satisfies the monadic descent conditions [6, Prop. 4.7.3.16].

Therefore the BBL comparison is an equivalence iff (v) holds, iff $\mathcal{A}$ is chirally Koszul.

4.6. **Kac–Shapovalov determinant.**

(i) ⇒ (ix). If $\mathcal{A}$ is chirally Koszul, the PBW filtration is strict on the vacuum module: $\mathrm{gr}^F M_0(\mathcal{A}) \cong \mathrm{Sym}^{\mathrm{ch}}(V)$ where $V = F_1\mathcal{A}/F_0\mathcal{A}$. Strictness of the PBW map implies that the Shapovalov form G_h is non-degenerate in the bar-relevant range.

(ix) $\Rightarrow$ (i). Conversely, non-degeneracy of G_h in the bar-relevant range means the PBW-to-bar comparison map $\iota\colon \mathrm{Sym}^{\mathrm{ch}}(V)_h \to \bar{B}_h(\mathcal{A})$ is injective at every bar-relevant weight h. Suppose for contradiction $d_r \neq 0$ for some $r \geq 2$. Then there exists $x \in E_r^{p,q}$ with $d_r(x) \neq 0$; let $\tilde{x} \in F^p\bar{B}$ be a lift. If $\tilde{x} \in \mathrm{im}(\iota)$, it represents a nonzero class in $\mathrm{gr}^p(\bar{B}) \cong \mathrm{Sym}^p(V)$, and $d_r(\tilde{x})$ represents a nonzero class in gr^{p+r}. By PBW strictness (injectivity of ι), this contradicts the acyclicity of $\mathrm{Sym}(V)$ as a Koszul complex. Hence $d_r = 0$ for all $r \geq 2$, giving (ii), hence (i).

4.7. **The Lagrangian criterion, conditionally.**

(i) $\Rightarrow$ (xi). Under the perfectness and non-degeneracy hypotheses on the ambient tangent complex [10, Prop. 6.3]: on the Koszul locus, the bar-cobar adjunction provides a free resolution of $\mathcal{A}$, and the complementarity splitting

$$\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!) \;\simeq\; H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$$

identifies $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ as complementary subspaces of $\mathcal{M}_{\mathrm{comp}}$. The (-1)-shifted symplectic structure on $\mathcal{M}_{\mathrm{comp}}$ makes each of these subspaces isotropic; complementarity gives maximal dimension, hence both are Lagrangian.

(xi) $\Rightarrow$ (i). If $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians, then their derived intersection

$$\mathcal{M}_{\mathcal{A}} \times^h_{\mathcal{M}_{\mathrm{comp}}} \mathcal{M}_{\mathcal{A}^!}$$

is discrete: transverse Lagrangian intersection in a (-1)-shifted symplectic space has expected dimension 0. The derived intersection computes the twisted tensor product $K_\tau(\mathcal{A}, \mathcal{A}^!)$; its discreteness is exactly acyclicity, which is Koszulness (Definition 2.3).

4.8. **$\mathcal{D}$-module purity, one direction.**

(xii) $\Rightarrow$ (x). Suppose each $\bar{B}^{\mathrm{ch}}_n(\mathcal{A})$ is pure of weight n as a mixed Hodge module. By Saito's theory [14], the weight filtration on a pure Hodge module splits up to semisimplification; for the bar complex, this splitting implies that the restriction $i_S^!\,\bar{B}_n(\mathcal{A})$ to any FM boundary stratum S is again pure. Purity plus the alignment of the characteristic variety to FM strata forces acyclicity outside degree 0 on every stratum, which is condition (x).

Remark 4.1 (Converse direction). The converse, that Koszulness implies $\mathcal{D}$-module purity of the bar complex, is established for the affine Kac–Moody family in [10, Prop. 7.5]: there the bar complex is identified with a geometric complex whose purity is known via the Riemann hypothesis for weight filtrations. For Virasoro and the $\mathcal{W}$ algebras, the bar complex does not have an obvious geometric origin of this form, and the converse is open.

This completes the proof of Theorem 3.1.

5. Examples

We verify the meta-theorem on four standard families. In each case we identify the condition that is easiest to verify, and use the theorem to propagate it to the others.

5.1. **Heisenberg.** The Heisenberg vertex algebra Heis_k at level k is generated by a single current $J(z)$ with OPE

$$J(z)J(w) \;\sim\; \frac{k}{(z-w)^2}.$$

At nonzero level, the PBW filtration is strict: the associated graded is a free polynomial Poisson vertex algebra. The Kac–Shapovalov determinant is a product of factors $(n-h)^{\mu_n}$ for the weight lattice, and is nonzero generically. Condition (ix) holds, and by the meta-theorem so do all the other conditions.

In the shadow tower language, Heis_k is a Gaussian algebra (depth $r_{\max} = 2$): the bar complex is abelian and the chiral Hochschild cohomology is concentrated in the single bar degree $\{0, 1, 2\}$ that is visible on any free theory. The A_∞-structure is strictly associative, and the bar–cobar counit is a quasi-isomorphism by direct computation.

Remark 5.1. The level-k formulas are consistent with the vanishing rule at $k = 0$: the relevant classical r-matrix is $r(z) = k\,\Omega/z$, which vanishes at $k = 0$. At nonzero level, Heisenberg is chirally Koszul by all ten criteria; the shadow constant is $\kappa^{\mathrm{Heis}} = k$.

5.2. **Affine Kac–Moody at generic level.** Let $\hat{\mathfrak{g}}_k$ be the affine Kac–Moody vertex algebra of a simple Lie algebra $\mathfrak{g}$ at level k. The Kac–Shapovalov determinant is given by the Kac determinant formula; it vanishes precisely at the Shapovalov walls $k + h^\vee = p/q$ for a discrete set of rationals depending on the weight. Away from these walls, condition (ix) holds, and the meta-theorem gives chiral Koszulness.

For the bar–cobar pair $(\hat{\mathfrak{g}}_k, \hat{\mathfrak{g}}_{k'})$ with $k + k' = -2h^\vee$ (the Killing self-duality), the shadow constant κ is

$$\kappa^{\mathrm{KM}}(\mathfrak{g}, k) \;=\; \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}.$$

At the critical level $k = -h^\vee$, the shadow constant κ^{KM} vanishes; the Shapovalov determinant acquires zeros of the corresponding order, and the Kac–Shapovalov input (condition (ix)) enters its degenerate regime. The critical level lies outside the generic Koszul locus treated here. At generic $k \neq -h^\vee$, $\kappa^{\mathrm{KM}} \neq 0$ and the meta-theorem closes.

As a numerical sanity check, substitute $\mathfrak{g} = \mathfrak{sl}_2$ (so $\dim \mathfrak{g} = 3$, $h^\vee = 2$) at level $k = 1$:

$$\kappa^{\mathrm{KM}} = \frac{3(1+2)}{2 \cdot 2} = \frac{9}{4}.$$

This agrees with the direct computation of the second shadow coefficient from the OPE [10, §3.1], and at $k = -h^\vee = -2$ one recovers $\kappa^{\mathrm{KM}} = 0$, reproducing the critical-level degeneration. At the trivial level $k = 0$, the classical r-matrix $r(z) = k\,\Omega/z$ vanishes and $\hat{\mathfrak{g}}_0$ reduces to an abelian chiral algebra; this abelian limit remains chirally Koszul (it is Gaussian in the shadow-depth sense), but it lies outside the generic level locus on which the Kac–Shapovalov input of condition (ix) is most informative.

5.3. **Virasoro in the Koszul locus.** The Virasoro vertex algebra Vir_c at central charge c has shadow constant $\kappa^{\mathrm{Vir}} = c/2$. The Kac determinant formula for Vir_c gives the discrete set of degenerate central charges $c_{p,q} = 1 - 6(p-q)^2/(pq)$; away from these, the vacuum module is irreducible in the bar-relevant range, and condition (ix) holds.

At $c = 13$, the Koszul pair $(\mathrm{Vir}_c, \mathrm{Vir}_{26-c})$ is self-dual, and the duality constant $c + c' = 26$ (the Koszul conductor of Virasoro) reproduces the critical dimension of the bosonic string. At these generic values, Vir_c is chirally Koszul and the meta-theorem closes.

The multi-weight obstruction for Virasoro at genus $g \geq 2$ enters Theorem H through a correction $\delta F_g^{\mathrm{cross}}$ in the multi-weight regime. This does not affect chiral Koszulness per se, but modifies the shadow tower structure that is visible in condition (viii) at higher genus.

5.4. **The $\beta\gamma$ system.** The $\beta\gamma$ system is a free-field vertex algebra with commuting fields $\beta(z), \gamma(z)$ satisfying $\beta(z)\gamma(w) \sim 1/(z-w)$. Its associated graded is the Poisson vertex algebra on $T^*(\mathcal{O}_X)$ with the canonical bracket; the bar complex is an exterior algebra on cotangent generators.

The A_∞-structure is strictly formal: all higher operations $m_n = 0$ for $n \geq 3$, since the OPE has no quartic or higher poles. Condition (iii) is manifest, and the meta-theorem closes. In shadow-tower language, the $\beta\gamma$ system is a contact algebra (depth $r_{\max} = 4$): the shadow tower is algebraic of degree 2 with nonzero quartic seed.

5.5. **Summary.**

Family	Easiest input	Propagated by
Heisenberg	(ix) det $\neq 0$	core circuit
Affine Kac–Moody	(ix) Kac det	core circuit
Virasoro (Koszul locus)	(ii) PBW collapse	(ii)$\Leftrightarrow$(v)
$\beta\gamma$ system	(iii) A_∞ formality	(iii)$\Leftrightarrow$(ii)

In each case the verification requires a single condition, and the meta-theorem converts it to all eight genuinely independent bidirectional equivalences (among (i)–(x), with (vi) a proved consequence of (v) and (viii) a one-way chiral Hochschild consequence on the Koszul locus) at once. The Lagrangian criterion (xi) holds unconditionally for Heisenberg and for Kac–Moody at generic level (the perfectness hypothesis is satisfied). $\mathcal{D}$-module purity (xii) is proved for the Kac–Moody family and open for Virasoro and $\mathcal{W}$.

6. Open questions

6.1. **The converse of $\mathcal{D}$-module purity.** The only condition whose equivalence with the rest is not fully established is (xii). The implication (xii) $\Rightarrow$ (x) is unconditional (Section 4); the converse is known for the affine Kac–Moody family (Remark 4.1) but open for Virasoro and $\mathcal{W}$.

Conjecture 6.1. *For every chirally Koszul chiral algebra $\mathcal{A}$ on a smooth projective curve, each $\bar{B}_n^{\mathrm{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to the FM boundary strata.*

The obstacle is that the bar complex of Virasoro does not have an obvious geometric origin from which purity could be read off; any proof of the converse must proceed either by a direct computation of the weight filtration on $\bar{B}_n^{\mathrm{ch}}(\mathrm{Vir}_c)$, or by an indirect route via a geometric model (e.g., a resolution of the Virasoro bar complex by a pure complex on a related moduli space).

6.2. **The perfectness hypothesis for (xi).** Condition (xi) requires perfectness and non-degeneracy of the ambient tangent complex, which is verified for the standard landscape [10, Prop. 6.3] but not known in full generality. A natural question is whether perfectness holds for every chirally Koszul algebra.

Conjecture 6.2. *If $\mathcal{A}$ is chirally Koszul, the ambient tangent complex at $\mathcal{A}$ in $\mathcal{M}_{\mathrm{comp}}$ is perfect of tor-amplitude $[0, 2]$.*

If this conjecture holds, the Lagrangian criterion (xi) is unconditionally equivalent to the other ten.

6.3. **The Koszul locus beyond the standard landscape.** A natural family of chiral algebras outside the standard landscape consists of the non-principal $\mathcal{W}$-algebras of type ADE. The Koszul duality for hook-type $\mathcal{W}$-algebras of type A has been established in [10]; other nilpotent orbits remain open and are controlled by the associated variety hierarchy recorded in [10].

6.4. **Shadow depth and Koszulness.** The shadow depth classification (Gaussian, Lie, contact, mixed) and chiral Koszulness are logically distinct: Koszulness concerns the diagonal concentration of the bar cohomology, while shadow depth concerns the degree at which the shadow tower terminates. All standard families are chirally Koszul (generic level), but only Gaussian (Heisenberg) and contact ($\beta\gamma$) have finite shadow depth; Virasoro and $\mathcal{W}$-algebras have $r_{\max} = \infty$ (class M). The meta-theorem does not distinguish between these.

Remark 6.3 (Koszulness is not SC formality). A subtlety worth emphasizing: chiral Koszulness is an assertion about H^* in diagonal bidegree; it is *not* the same as Swiss-cheese formality (which is the vanishing of higher operations on the open-closed bar complex). All standard families are chirally Koszul;

only a narrower class (class G) is Swiss-cheese formal. In this paper we have focused on chiral Koszulness.

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Appendix A. Note added in proof: 2026-04-16/17 strengthenings

Between the first circulation of this note and the 2026-04-17 Beilinson-rectification audit, the twelve-equivalence meta-theorem acquired a sharper internal structure at genus 0 and refined scope qualifiers at higher genus.

Seven-fold tfae hub-and-spoke at genus 0**.** The core circuit (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (v) $\Leftrightarrow$ (viii) of Theorem 3.1 is reorganised, at genus 0, as a hub-and-spoke TFAE (Vol I `chapters/theory/ftm_seven_fold_tfae_platonic.tex`, `thm:ftm-seven-fold-tfae-via-hub-spoke`). Each spoke is bidirected to the PBW E_2-collapse *hub*:

(S1) Koszul morphism (condition (i));
(S2) bar-cobar counit quasi-isomorphism (condition (v));
(S3) bar-cobar unit weak equivalence;
(S4) twisted tensor product acyclicity (equivalent to PBW);
(S5) bar cohomology concentrated in weight 1 (the Priddy condition, condition (ii));
(S6) Swiss-cheese formality (class-G-scoped: equivalent to the hub on the class-G stratum; class-L, C, M members of the Koszul locus are Koszul but not SC-formal);
(S7) ChirHoch* polynomial in $\{0, 1, 2\}$ (condition (viii)).

Spokes (S1)–(S5) and (S7) are hub-equivalent on the full Koszul locus; Spoke (S6) is hub-equivalent only on class G. Pairwise bidirection count on the Koszul locus reduces to six proved spokes; the remaining equivalences follow by transitivity. Spoke (S4) is the load-bearing non-tautology, witnessed on $V_k(\mathfrak{sl}_2)$ at generic level.

Scope qualifiers. (N1-Q1) Condition (vi) Barr–Beck–Lurie monadicity. The monadicity equivalence is stated at genus 0 level; the all-genera extension is contingent on the Francis–Gaitsgory $(\infty, 2)$-descent and inherits the scope of Theorem $\mathrm{A}^{\infty,2}$ (`thm:A-infinity-2`, Vol I `chapters/theory/theorem_A_infinity_2.tex`).

(N1-Q2) Condition (vii) factorization homology. The concentration-in-degree-0 statement of (vii) is uniform-weight at $g \geq 2$; multi-weight families ($\mathcal{W}_N$, $N \geq 3$) carry a cross-channel correction $\delta F_g^{\mathrm{cross}}$ at $g \geq 2$, explicit at $(W_3, g = 2)$: $\delta F_2^{\mathrm{cross}}(W_3) = (c + 204)/(16c)$ (`prop:delta-f-cross-w3-g2`). The meta-theorem (vii) is stated on the uniform-weight lane; outside the uniform-weight lane the equivalence with (i) still holds, with a multi-weight interpretation of the concentration.

(N1-Q3) Condition (viii) ChirHoch* **concentration.** The Step 3 circularity risk in the original proof of Theorem H, used at the (v) $\Rightarrow$ (viii) and (viii) $\Rightarrow$ (v) arrows, is rerouted through `thm:pbw-koszulness-criterion` (Vol I `chapters/theory/chiral_pbw_koszulness.tex`): the Shelton–Yuzvinsky Koszulity of braid-arrangement Orlik–Solomon algebras plus the chiral quadratic-Koszul lemma make bar-concentration and FM-tower collapse PARALLEL consequences of PBW, not sequential, retiring the earlier sequential routing.

(N1-Q4) Condition (xi) Lagrangian criterion. Perfectness, previously a hypothesis, is now a theorem on the standard landscape (`prop:perfectness-standard-landscap` verified family-by-family: Heisenberg, affine Kac–Moody at non-critical level via Kac–Kazhdan, Virasoro at generic c, $\mathcal{W}_N$ via Fateev–Lukyanov, lattice via theta/eta, $\beta\gamma$ via character. Consequently (xi) becomes an UNCONDITIONAL equivalence on the standard landscape; off-landscape, it remains conditional on perfectness.

(N1-Q5) Condition (xii) $\mathcal{D}$-module purity. One direction ((xii)$\Rightarrow$(x)) remains PROVED; the converse (x)$\Rightarrow$(xii) was proved in Vol I for the affine Kac–Moody family; it is not extended in the 2026-04-16 wave and remains open in general.

GRT_1**-torsor structure: heptagon closure.** The five presentations of the Koszulness moduli are organised, in Vol II `chapters/theory/sc_chtop_heptagon.tex` (`thm:sc-chtop-heptagon`), as a heptagon of seven named edge theorems: five classical faces plus face (6) Drinfeld-centre $Z(\mathrm{Rep}_{\mathrm{fact}}(\mathcal{A})) \simeq \mathrm{Rep}_{\mathrm{fact}}(Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}))^{E_2}$ via categorified bar-cobar with half-braiding, plus face (7) derived-algebraic-geometry via PTVV on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, B\mathrm{SC}\text{-Alg})$. The GRT_1-torsor of this paper is the automorphism torsor of this heptagon; the chart atlas of Theorem 1.2 is compatible with the heptagon's seven-fold global structure.

Admissible Kazhdan–Lusztig periodic CDG. For a chiral algebra at admissible level $k+h^\vee = p/q$, the meta-theorem extends via the periodic CDG filtration $F^n = \ker(Q^n_{\mathrm{adm}})$ on $\mathrm{KL}^{\mathrm{adm}}_k$ (`thm:periodic-cdg-is-koszul-compatible`, Vol I `chapters/theory/periodic_cdg_admissible.tex`): periodic CDG is Koszul-duality-compatible, and the bar-cobar adjunction $\Omega^{\mathrm{ch}} \dashv B^{\mathrm{ch}}$ descends UNCONDITIONALLY to $\mathrm{KL}^{\mathrm{adm}}_k \rightleftarrows (\mathrm{KL}^{\mathrm{adm}}_{k^!})^{\mathrm{op}}$. The class-$\mathsf{M}$ admissible minimal model $\mathrm{KL}^{\mathrm{adm}}_{\mathrm{Vir}}(c_{p,q})$ has $(p-1)(q-1)/2$ simples, all finite-length (`cor:class-M-admissible-minimal-model`). This closes the admissible-level sector of the meta-theorem's coverage on the standard landscape.

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